Cohort analysis

Week 4, Lecture 2 — B2C to $10k MRR

B2C to $10k MRR: A 10-Week Growth Manual · Autumn 2026

What we will cover

- Building a cohort retention table from scratch in SQL
- Reading the diagonal: which cohort changed shape and why
- Power-user analysis: who are your top 5%, and what did they do differently
- Churn diagnosis: involuntary (card decline) vs voluntary (lost interest)

Building the cohort table

Why cohort vs aggregate matters

- An aggregate retention number averages over all cohorts
- A new product can look healthy in aggregate while every individual cohort is decaying
- This happens when newer (worse) cohorts are too small to drag the average down yet
- Berezovsky (2022): retention is an output metric; measuring it in aggregate hides the signal

The core SQL pattern

```sql
WITH signups AS (
  SELECT user_id,
         DATE_TRUNC('week', created_at) AS signup_week
  FROM events
  WHERE event_name = 'ACTIVATION_EVENT'
),
activity AS (
  SELECT user_id,
         DATE_TRUNC('week', created_at) AS activity_date
  FROM events
  WHERE event_name = 'RETENTION_EVENT'
)
SELECT
  s.signup_week,
  COUNT(DISTINCT s.user_id)                         AS cohort_size,
  COUNT(DISTINCT CASE
    WHEN a.activity_date = s.signup_week + INTERVAL '1 day'
    THEN a.user_id END) * 100.0 / COUNT(DISTINCT s.user_id) AS d1_pct,
  COUNT(DISTINCT CASE
    WHEN a.activity_date = s.signup_week + INTERVAL '7 days'
    THEN a.user_id END) * 100.0 / COUNT(DISTINCT s.user_id) AS d7_pct,
  COUNT(DISTINCT CASE
    WHEN a.activity_date = s.signup_week + INTERVAL '30 days'
    THEN a.user_id END) * 100.0 / COUNT(DISTINCT s.user_id) AS d30_pct
FROM signups s
LEFT JOIN activity a ON s.user_id = a.user_id
GROUP BY s.signup_week
ORDER BY s.signup_week;
```

What the query produces

```
signup_week   cohort_size   d1_pct   d7_pct   d30_pct
2026-07-28         42         61%      34%      19%
2026-08-04         58         63%      36%      21%
2026-08-11         71         64%      38%      23%
2026-08-18         85         62%      41%      (too early)
```

Read each row as one cohort. D30 for the most recent week is always incomplete.
Mark it "too early" rather than reporting a misleadingly low number.

Reading the diagonal

The diagonal reveals when something changed

- In a cohort table, cells along the same calendar date form a diagonal
- If D7 jumps for every cohort that signed up after a specific date, something changed that date
- Could be a product release, a pricing change, a new acquisition channel, or a bug fix
- The diagonal is your most precise instrument for attributing product changes to retention

How to read the diagonal in practice

```
signup_week   d7_pct   d30_pct
2026-07-07     28%       12%     <- before change
2026-07-14     27%       11%     <- before change
2026-07-21     38%       22%     <- jump: something shipped
2026-07-28     40%       24%     <- persists
```

Check your deploy log for the week of July 21.

One change (onboarding rewrite, email sequence, paywall timing) produced this jump.

Power-user analysis

Who are your top 5%?

- Power users are not the users who pay the most: they are the users who use the product most
- Define by an event count threshold: "users with more than N events in the past 30 days"
- Or by a specific action: "users who completed the core workflow at least once per week"
- The threshold should map to your north-star metric (activation-rate, north-star-metric)

The power-user query

```sql
SELECT
  user_id,
  COUNT(*) AS event_count_30d
FROM events
WHERE event_name = 'RETENTION_EVENT'
  AND created_at >= NOW() - INTERVAL '30 days'
GROUP BY user_id
ORDER BY event_count_30d DESC
LIMIT (SELECT CEIL(COUNT(DISTINCT user_id) * 0.05) FROM events
       WHERE created_at >= NOW() - INTERVAL '30 days');
```

Then: what did these users do in their first 7 days that others did not?

What to look for in the week-1 path

- Did power users complete a specific action that most users skipped?
- Did they invite a second user, connect an integration, or create a second item?
- Did they receive a specific email or notification that other users did not?
- That action is your activation intervention target for week 5

Churn diagnosis

Two kinds of churn, two different fixes

- **Voluntary churn:** user consciously cancelled. Cause: insufficient value, better alternative, cost concerns.
- **Involuntary churn:** payment failed. Cause: expired card, insufficient funds, bank block.
- RevenueCat (Tideman, 2025): involuntary churn is 28% of total churn on Google Play, 14% on iOS
- Baremetrics (Jackson, 2020): involuntary churn costs ~9% of MRR monthly; median 410% ROI on dunning

Top reasons users cancel (Tideman, RevenueCat 2025)

```
Reason                          Share of churn
Insufficient usage                 37%
Cost concerns                      35%
Billing errors (involuntary) up to 28% (Google Play) / 14% (iOS)
Found better app                    9.5%
Technical issues                    7%
```

"Insufficient usage" is a retention problem, not a price problem.
Fix D7 before adjusting price.

Involuntary churn: the fix stack

1. **Pre-dunning email:** warn 7 days before card expiry
2. **Smart retry logic:** retry failed payments at different times (not same time next day)
3. **Grace period:** keep access for 3-7 days after failure while retrying
4. **Multi-channel notification:** email + in-app + SMS for high-value customers
5. **Card update prompt:** deep-link directly to payment update screen

Google Play users churn at 2x the iOS rate from billing alone. If your app is cross-platform, fix Android dunning first.

> "Retention is the most important metric for long-term business health and a flattening retention curve is the core signal of product-market fit."

<footer>Lenny Rachitsky, "What is good retention?", 2020</footer>

Take-aways

1. Cohort tables reveal what aggregate numbers hide: which signup week changed shape and when.
2. Read the diagonal to attribute product changes to retention outcomes.
3. Power users are defined by behavior (event count, specific action), not demographics.
4. Involuntary churn is 14-28% of your total churn and recoverable with dunning logic alone.
5. Diagnose voluntary vs involuntary before deciding whether to ship features or fix billing.

What is next

- **Section (this week):** run the cohort query against your own data; name the curve shape
- **Week 4 reading:** retention theory, the Hook Model, case study on Duolingo streaks
- **Week 5 (next week):** mid-course pivot/persevere check; acquisition channels
- **HW 2 due:** pricing and paywall (check the syllabus for submission details)